From AGI-Native 6G Networks to Active Inference: Are Psychoactives the New Frontier of AGI?

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***Abstract*—Future artificial general intelligence (AGI)-native 6G networks will widen research from a narrow computer science perspective to a broader neuroscience as well as robotics perspective, given that key cognitive abilities of AGI such as lifelong learning (LL) via environmental coupling have been studied in both fields for decades. After highlighting key lessons from neuroscience and robotics with regard to more advanced brain models and LL capabilities, this paper focuses on active inference, an ideal methodology for advancing AGI, with its recurrent dark-room problem and how to resolve it by applying biomimetic psychoactives in the future 6G World Brain. We show that active inference points in the exactly opposite direction of today's envisioned AGI-native 6G trajectory. The journey beyond 6G might not be outward via AI embodiment, but inward into what neuroscientists coined the brain's dark energy, thereby resolving the dark-room problem and helping us penetrate ever deeper into cognitive space.**

I. Introduction

FUTURE 6G networks will sit at the nexus of communication, computation as well as intelligentization and robotization, especially given the advent of AI robots [1]. Recently, Yann LeCun has left Meta to pursue research on advanced machine intelligence (AMI) that replicates how a child or a baby animal learns, i.e., how biological agents learn. There is now a growing awareness among leading artificial intelligence (AI) researchers about the importance of bridging AI and neuroscience, leading to the development of biologically informed AI models [2]. Towards this end, recent initiatives have been launched such as the 2024 NeurIPS Workshop on "NeuroAI: Fusing Neuroscience and AI for Intelligent Solutions," including a panel discussion between two of the biggest names in AI today: Yoshua Bengio, a pioneer in deep learning, and Karl Friston, the father of *active inference* and main topic of this paper.

With the emergence of artificial general intelligence (AGI)-native 6G networks, *AI embodiment* will allow AI agents to transcend their purely computational nature and enable them to interact with their surrounding physical environment via sensory grounding through 6G, giving rise to novel types of agent-environment interaction, most notably human-AI interaction. To realize the vision of AI embodiment through 6G, the authors of [3] argue that reinforcement learning (RL) with human feedback is essential for embodied AI agents to be grounded in the environment, though it is debatable whether the reward of RL is enough to do so, and that advanced capabilities such as lifelong learning (LL) with reduced training overhead are key in AGI. However, while LL is certainly key to advancing AI embodiment, it is worth mentioning that it has been studied for decades since the 1990s in other areas, most notably robotics [4]. Clearly, there might be some lessons to be learned from *lifelong robot learning* and adopt them to AI embodiment (to be further discussed below).

Recently, an excellent research outlook on the evolution of future AGI-native networks in the beyond 6G era was presented in [5]. The authors advocate to fundamentally revisit the 6G concept of AI-native wireless systems, given that they rely mostly on conventional AI tools that struggle to cope with complex real-world environments. They tackle the fundamental question of how to design AGI-native 6G networks with new cognitive abilities such as common sense, a cognitive trait common to all humans. Their proposed vision foresees a new breed of AGI-native 6G networks operating with an *AGI-native telecom brain* that is premised on Yann LeCun's cognitive brain architecture composed of different modules representing various cognitive abilities such as perception, action planning, short-term memory, and causal world model.

At the downside, the statement in [5] that "remarkably to date, there are no rigorous world models that can permit autonomous agents to manipulate representations so that they can predict the future" might be true for the field of computer science exemplified by Yann LeCun, a worldwide renown computer scientist and pioneer of deep learning (DL), but certainly not for neuroscience and in particular active inference – a unified theory of mind, brain, and behavior pioneered by Karl Friston, the world's most frequently cited neuroscientist [6]. Active inference is a biomimetic mathematical framework that is premised on first principles of statistical physics found in self-organizing/evolving complex adaptive systems. It reconciles and unites predictive, enactive, and cybernetic theories. As we shall see, active inference is an ideal methodology for developing advanced AGI systems, while overcoming the limitations of today's AI related to LL, among others. Obviously, similar to robotics, there might be some lessons to be learned by widening the scope from a narrow computer science perspective to a broader neuroscience one, especially in light of the recent illuminating panel discussion between Yann LeCun and Karl Friston at the World Economic Forum on *what AI is today, and what it can become* [7], as further elaborated on below.

This paper builds on our latest work titled "From Artificial Intelligence to Active Inference: The Key to True AI and the 6G World Brain," in which we've cited a widely read WIRED article whose closing sentence states that, in the end, the acronym AI might actually not stand for artificial intelligence,

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but active inference – the key to true AI [8]. The goal of this paper is as follows. After briefly reviewing the main premises of [8], we will delve into active inference's *dark-room problem*, a recurrent puzzle faced by biological agents like us. We advance active inference by widening its scope from biological to psychoactive agents, or *psychoactives* for short, which hold great promise to induce profound shifts in perception and cognition of the future 6G World Brain by harnessing their "door in the wall."

The remainder of the paper is structured as follows. Section II highlights some of the key lessons computer science may learn from neuroscience and robotics with regard to more advanced brain models and LL capabilities. Section III further elaborates on the aforementioned exchange between LeCun and Friston on active inference, paying particular attention to its central *free-energy principle* – a unifying theory of brain function – and how free-energy minimization leads to the dark-room problem and how to resolve it via psychoactives. After describing our vision of the 6G World Brain[1] premised on active inference in more detail in Section IV, Section V explores the beneficial impact of psychoactives on AGI models. Finally, Section VI concludes the paper.

II. Brain Models and Lifelong Learning: Computer Science vs. Neuroscience & Robotics

For decades, it has been the dominant metaphor to view the brain as a computer, drawing analogies from computer science such as that the brain processes input information and then makes a decision. This old "outside-in" strategy of the brain is now replaced with a new "inside-out" paradigm in modern neuroscience, where the *brain does not process information: it creates it* [9]. In this new neuroscience dictum, the brain acts as a probabilistic prediction machine that predicts sensory inputs by means of action-oriented predictive processing. More specifically, the brain uses a generative model to generate actions and predicts the sensory consequences in an "inside-out" manner of *embodied action*, while processing the actual sensory inputs as prediction errors. Or, put differently, the predictive brain creates a "realm of possibilities" (i.e., hallucinations) to perform a "preplay of the future" and then uses perception to control its hallucinations via action-based grounding.

It's this embodied action how robots continuously learn through a process known as lifelong learning. Unlike AI, research on lifelong robot learning has already started decades ago in the 1990s [4]. Learning robots can operate in initially unknown environments and compensate for changes, since they can re-adjust their internal belief about the environment and themselves grounded in the real world. To do so, autonomous robots embedded into a lifelong learning context perceive the state of their environment through sensors and act using effectors, giving rise to the *action-perception loop* that is instrumental to any form of lifelong learning in both machines and living organisms. However, the authors of [4] also pose the question that is of central importance to AI researchers still today: Why do natural agents such as humans learn so much better than artificial agents?

And this is where active inference comes into play, which studies how processes of action, perception, learning, planning, and goal-directed behavior emerge from active inference *biological agents*, including but not limited to humans, premised on the so-called Hamiltonian principle of least action, which facilitates the most energy-efficient form of learning with no big data requirement necessary for training. The Hamiltonian principle of least action is a first principle that connects active inference to the domains of statistical physics, thermodynamics, and nonequilibrium steady states [6]. More importantly, the physics underlying active inference are *pointing in the exactly opposite direction of today's AI trajectory*. From the point of view of the physicist, one should be looking for the least data instead of big data in order to resolve uncertainty, i.e., learn in a more intelligent way [10]. With this being said, let's explore active inference in more detail next.

III. Active Inference – The Key to True AI: Less Artificial, More Intelligent

As briefly mentioned in Section I, in a panel at the 2024 World Economic Forum, two of the biggest names in AI – computer scientist Yann LeCun and neuroscientist Karl Friston – agreed that today's AI is profoundly lacking the key elements found in living intelligent systems in terms of perception, memory, reasoning, and generating actions [7]. Where they differ was in how to get there. LeCun, a pioneer in DL, argues that we don't know how to do learning or train systems any other way than DL, an engineering approach to AI invented decades ago. Conversely, Friston, the father of active inference, challenges the reliance on DL by advocating a new and better alternative, rooted in the *free-energy principle*, a unifying theory of brain function explaining how neurons and biological systems in general learn, adapt, and self-evolve in nature.

A. The Free-Energy Principle

The free-energy principle (FEP) is essentially a mathematical formulation of how biological systems resist a natural tendency to disorder in the face of a constantly changing environment [11]. The FEP provides the answer to a fundamental question: How do self-organizing adaptive systems avoid surprising states for the sake of their survival? They can do this by minimizing their free energy, which can be shown to be an upper bound on the surprise of sensory states. This means that the probability distribution of sensory states must have low *entropy*, a measure of uncertainty denoting the average surprise (i.e., there is a high probability that a biological system will be in any of a small number of states, and a low probability that it will be in the remaining states).

All the heavy lifting in active inference is done by minimizing free energy, which comes in two flavors: *Variational free energy* F and *expected free energy* G. It's crucial to understand the role of variational and expected free energy in active inference. Simply put, F and G provide an active inference agent's retrospective view ("how good has this action

[1]We note that the term *6G World Brain* pays tribute to the fact that AGI-native 6G is anticipated to play a significant role in advancing Nikola Tesla's prophecy that "when wireless is perfectly applied, the whole Earth will be converted into a huge brain."

plan turned out so far?") and prospective view ("how good do I expect things to go if I continue to follow this action plan") on pursuing a certain policy (i.e., sequence of actions), respectively. More formally, an active inference agent aims at finding beliefs in its generative model for which observations provide the most evidence, a process known as *self-evidencing*. Importantly, the underlying generative model enables active inference agents to learn the causal structures of their world instead of merely correlations, endowing them with a *causal world model*, a capability largely absent in alternative AIs (e.g., RL). For further information on the FEP and active inference, we refer the interested reader to [12] for a recent overview of the history and future of active inference. Next, we turn to active inference's dark-room problem, the main focus of this paper.

B. The Dark-Room Problem

Although there is wide consensus about the general idea of active inference and its overarching FEP of surprise minimization, there is also a recurrent puzzle known as the *dark-room problem* [13]. Biological agents do not simply minimize surprise by taking up a position in the nearest dark room – i.e., seeking the least stimulating environment that affords low levels of surprise – even though this would be the best way to reduce surprise. Clearly, there seems to be something wrong, especially given that play, exploration, epistemic foraging, and creativity are core features of life. To resolve the dark-room problem, Karl Friston *et al.* argue in [13] whether active inference needs to be "part of some larger explanatory package" that involves principles other than minimization of surprise alone.

Recently, in [14], Karl Friston *et al.* proposed a solution referred to as the *enlightened room problem*, which lies not (or not only) in the surprise-minimizing brain but in the environment itself. By constructing challenging outside worlds, i.e., external environments, by means of environmentally mediated means, the creativity of agents may be cultivated to seek out anything that is different from what they already know via the external action-perception loop.

It is important to note, however, that our interactions with the external environment are just a small part of the brain's activity. A great deal of meaningful brain activity, consuming 60-80% of the brain's overall energy, occurs when a person is sitting quietly in a chair and doing nothing at all. In fact, certain brain regions – known as the default mode network (DMN) in neuroscience – were shown to experience an increased level of activity when not carrying out a task. Surprisingly, it was shown that the energy consumed by the brain's default mode is about 20 times that used by the brain when it responds to external stimuli. This intrinsic activity – often called the *brain's dark energy* [15] – continues even when the brain is at rest, such as during sleep or under anesthesia.

Instead of requiring constantly renewed outside worlds, adding to the external distraction of our today's hypermediated life, the dark-room problem of active inference may be resolved by seeing it as an opportunity to dive into the brain's dark energy by intentionally disengaging from the external action-perception loop instead of further emphasizing it. Turning the attention from the outside environment to the *inside invironment* might be the only way out of the recurrent dark-room puzzle – perhaps even intended so by human evolution. As a result, this might lead to an *enlightened brain in lieu of enlightened room* (to be further discussed below). It is interesting to note that reducing outside external stimuli also allows participants to engage in their experience more fully during psychedelic therapy. In [16], it was shown that the therapeutic effects of the *psychoactive agent* psilocybin induces profound shifts in perception and cognition of the human brain, often lasting weeks, months, or even years. In fact, the aforementioned DMN was shown to be affected the most by psilocybin, making its connections more malleable. According to [16], this malleability (technically referred to as neuroplasticiy) is what neurotherapists are harnessing to "open the door to change."

C. The Door in the Wall of Psychoactives: From Biological Agents to Psychoactive Agents

Active inference aims at biomimicking the way *natural intelligence* works in order to bridge the gap between artificial and natural intelligence. In [17], Michael Pollan asks the provocative question how amazing it is that so many kinds of plants have hit upon the precise recipes for molecules that fit snugly into receptors in human brains and can pass as human neurotransmitters. According to Pollan, psychoactive agents produced by plants may be profiled as (i) sedative (e.g., opium), (ii) stimulant (e.g., caffeine), and (iii) hallucinogen (e.g., psilocybin). However, unlike their sedative and stimulant counterparts, hallucinogenic agents induce a state of mental plasticity, one in which you are highly open to learning new patterns of thoughts and behavior. In doing so, they crack open the door of perception and cognition and show what Pollan calls the "door in the wall." Walking thru this door in the wall expands our perception and cognition to see how things really are beyond the trickle of information limited to what is biologically useful for our survival. Or, as Pollan puts is, instead of alternate reality, it promises infinitely more of this one.[2]

In the remainder of the paper, we advance active inference by widening its scope from biological to psychoactive agents, or *psychoactives* for short, which hold great promise to induce profound shifts in perception and cognition of the future 6G World Brain by harnessing their effect of enhanced plasticity.

IV. Active Inference and 6G World Brain

A. Generative Model vs. Generative Process

At the heart of active inference lies the design of a *generative model*. Active inference can operate on different kinds of generative model. The big challenge of active inference is to specify the most appropriate form of the generative model

[2]Note that the ultimate aim of xAI's Grok, the presumably most intelligent model in the world, is also the inquiry about the nature of reality by achieving a deep understanding of and interaction with it (visit https://x.ai for details).

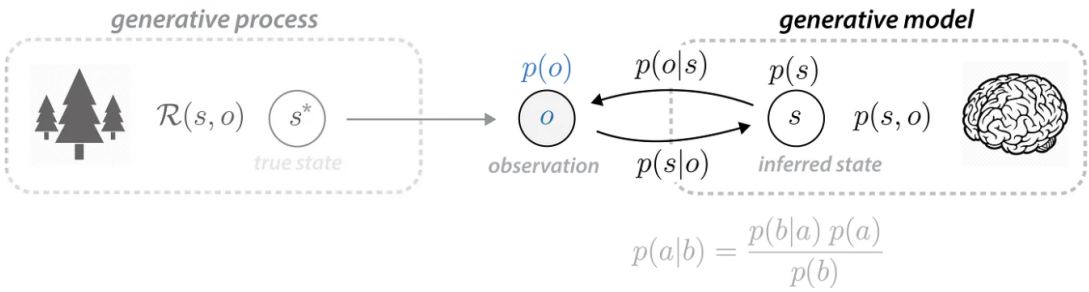

Fig. 1. At the heart of active inference lies the design of a generative model: Getting the probabilistic generative model $p(s, o)$ right for updating an active inference agent's belief about the true state s^* via Bayesian inference of state s (i.e., cause a in Bayes' theorem) from observation o (i.e., consequence b in Bayes' theorem) is the big challenge of active inference (Bayes' theorem is shown in light grey) [8].

for the problem at hand. Once this is appropriately formalized in terms of a generative model, the solution to the problem emerges under active inference [6].

In active inference, it is important to distinguish between the generative model and the *generative process* that describes the dynamics of the world external to the active inference agent. As illustrated in Fig. 1, the generative process corresponds to the process that determines the active inference agent's observation o. The agent can infer the hidden true state s^* from its observation o by using the well-known Bayes' theorem thru a process known as Bayesian inference (see details in Fig. 1). Belief update via Bayesian inference is where learning emerges. According to [6], in many practical applications, the active inference agent's generative model is assumed to closely biomimic the generative process that generates its observations, i.e., the *generative model biomimics the dynamics of the biological agent* under consideration.

In [8], our problem at hand was the 6G World Brain and how to realize it with active inference. In search for a suitable biological agent, we explored the role of Charles Darwin's "root-brain" hypothesis, which has been forgotten in the literature until 2005. The root-brain hypothesis has been slowly penetrating mainstream research since then, as recent advances in plant molecular biology unmasked plants as sensory and communicative organisms characterized by active, problem-solving behavior. It claims that the brain-like root apices monitor and integrate numerous parameters simultaneously and then translate these sensory experiences into complex motoric responses (similar to the action-perception loop in active inference). More importantly, the root-brain is made of the unique biological agent *mycorrhizal networks* found underground in all forests, which creates a symbiosis between underground fungal networks and interconnected trees. These mycorrhizal networks, representing the largest living organism on planet Earth, were highlighted by Karl Friston *et al.* as a prime example of active inference in their recent AGI vision and R&D roadmap for the next decade and beyond.

B. Modified OpenAI's Improved DDPM Generative Model

Denoising diffusion probabilistic models (DDPMs), using variational inference for training, are a class of latent variable models inspired by considerations from nonequilibrium thermodynamics [18], an inspiration grounded in physics they share with active inference. In [19], Prafulla Dhariwal and Alex Nichol from OpenAI showed that DDPMs are superior to generative adversarial networks (GANs) for image generation. Recently, we have shown in [8] that OpenAI's improved DDPM generative model can be modified such that it biomimics the following salient features of mycorrhizal networks:[3]

- **Exploration via Branching:** Mychorrhizal networks *branch* out into soil in an exploratory way by relying on fungal connections, resulting in increased disorder (i.e., entropy). These fungal connections are realized by fungi that interconnect the roots of different trees. We used the forward diffusion process of DDPM to biomimic exploration via branching, resulting in increased entropy.
- **Exploitation via Fusing:** Upon discovering resources, new fungal connections are *fused* that re-establish order (i.e., reverse-time or *negative* entropy). We used the backward denoising process of DDPM to biomimic exploitation via fusing, resulting in decreased entropy.
- **Embodiment via Entreating:** Fungi use mushrooms – fungi's fruiting bodies – to *entreat* the more-than-fungal world above ground. Mushrooms enable the embodiment of fungi to help them engage in agency. Specifically, they produce spores to disperse themselves in the real world.

In a nutshell, our modified DDPM model adapts and extends OpenAI's improved DDPM from an image generation context to the generation of small-world, scale-free network structures with adjacency tensors of increasing and decreasing entropy, respectively. We used NetworkX to generate a dataset consisting of 1,304,483 unique tensors, each defining a different small-world, scale-free synthetic network, for training OpenAI's improved DDPM (for further technical details, the interested reader is referred to [8]). As a result, our modified DDPM generative model is able to entreat the physical more-than-digital world and disperse itself in the real world via digital spores (to be explained below), thereby transforming DDPM into *embodied, enactive AI*.

Now, it is important to note that in their AGI vision and R&D roadmap, Friston *et al.* suggest that AGI will emerge from the interaction of intelligences networked into a cyber-physical ecosystem of artificial and natural intelligence, in which humans become integral participants via *human-AI*

[3] Source code available at: https://github.com/Nikahosseinii/genai-vs-active-inference/tree/main?tab=readme-ov-file

interaction. Human-AI interaction let humans interact with different types of AI, ranging from blockchain smart contracts operating on simple if-then programming to more advanced generative AI (GenAI) such as OpenAI's improved DDPM. According to MIT's Alex Pentland, humans can perceive things in a broadly competent way and thus have a common-sense understanding of the real world that they bring to most problems. *So what would happen if we replaced AI neurons with people?* Pentland calls this the *human AI strategy*, which will help humans cope with the emerging AGI.

In [8], we've applied Pentland's human AI strategy to our 6G World Brain, where traditional AI neurons are replaced with humans and active inference AGI agents, using 6G for their environmental coupling and sensory grounding. In our numerical work in [8], we considered the illustrative use case of embodied, enactive AI where our modified DDPM generative model performs the following two core functions: (i) *Structure*: Generative model dynamically couples pairs of humans via hallucinated scale-free networks with small-world properties that are widely found in biological, social, and man-made systems; and (ii) *Agency*: Similar to cross-pollination among trees, pairs of directly linked humans spread a desirable new social norm with probability $0 \leq r \leq 1$ incentivized by teleological tokens (i.e., purpose-driven blockchain tokens), which are dispersed like digital spores into the physical more-than-digital world. Our obtained results showed that the number of required time units it takes for spreading the new norm among all humans becomes as low as 2 time units due to the extremely large degree of network connectivity and nodal degree created by our modified DDPM, independently from whether r is small or large. At the downside, however, this comes at the expense of a huge number of actuated tokens on the order of 2,500 tokens ("token bill"), given that humans are heavily interconnected, thus engaging in way too many social norm spreading events than actually necessary.

C. POMDP-based Active Inference Generative Model

In active inference, generative models typically use the concept of *partially observable Markov decision process (POMDP)*, which offers a fairly expressive structure to model discrete state-space environments by means of parameters that can be expressed as tractable categorical distributions. A POMDP can be formally defined as a tuple of finite sets and matrices $(S, O, U, \mathbf{A}, \mathbf{B})$, where S denotes the set of states s causing observations $o \in O$ and U denotes the set of actions u exerted by the agent. Further, matrix $\mathbf{A}$ encodes the likelihood $p(o_\tau|s_\tau)$ at time τ, while matrix $\mathbf{B}$ encodes the one-step transition dynamics $p(s_t|s_{t-1}, u_{t-1})$ at time t. Importantly, instead of passively inferring what caused observations without exerting any actions on the environment, active inference agents actively infer by using the available actions in U of their POMDP-based generative model. Note that these actions stand for the term 'active' in active inference. Also note that apart from the aforementioned POMDP matrices $\mathbf{A}$ and $\mathbf{B}$, a generative model may comprise additional matrices, including matrix $\mathbf{C}$ for preferred outcomes, $\mathbf{D}$ for prior beliefs about each possible state at initial time point $\tau = 1$, and $\mathbf{E}$ for prior beliefs about habits.

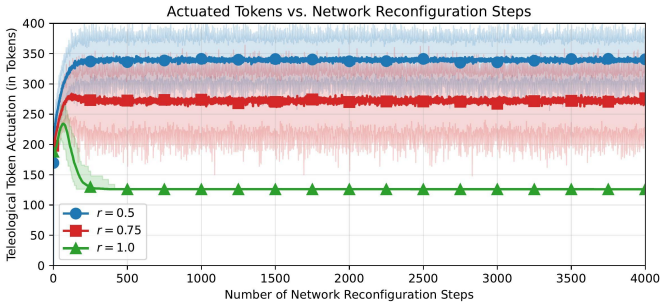

Fig. 2. Dynamics of our developed POMDP-based active inference generative model for different spreading probability $r \in \{0.5, 0.75, 1.0\}$: "Token bill" one order of magnitude lower than that of modified OpenAI's DDPM.

We have developed a POMDP-based active inference generative model biomimicking the generative process found in mycorrhizal networks. Our developed generative model got close to the minimum social contagion of 1 time unit, compared to 2 time units in the case of DDPM above. More importantly, Fig. 2 shows that the "token bill" of our proposed POMDP-based active inference generative model is about 250 tokens, i.e., *one order of magnitude lower* than that of DDPM. Note that Fig. 2 demonstrates a central cost vs. intelligence contrast between DDPM and active inference: While DDPM relies on indiscriminate noising, active inference follows a more intelligent belief-driven approach by adding only those edges that reduce the diffusion path, thus never wasting tokens on any redundant links. While the time units to spread measure the time it takes to move the collective of multiple agents from an old belief (i.e., social norm) to a new one, the "token bill" measures how *intelligently* this belief update is done.

V. AGI and Psychoactives: Expanding the Free-Energy Landscape

In this section, we aim at further enhancing our POMDP-based active inference generative model via psychoactives. Let our starting point be its inherent AGI capability of lifelong learning (LL), briefly mentioned in Section I. Figure 3 depicts the belief-update magnitude $\Delta_{\mathrm{KL}}(s)$, which measures the Kullback-Leibler (KL) divergence between the posterior belief $Q(s)$ and previous belief $Q(s-1)$ of a given agent at each network reconfiguration/rewiring step s, formally defined as:

$$\begin{aligned} \Delta\mathrm{KL}(s) &= \mathrm{KL}\big[\, Q(s) \,\|\, Q(s{-}1) \big] \\ &= \sum_{f \in \{\mathrm{norm},\, \mathrm{MB},\, \mathrm{net},\, \mathrm{tok}\}} \sum_{i} Q_{f,i}(s) \\ &\quad \times \ln\left(\frac{Q_{f,i}(s)}{Q_{f,i}(s-1)} \right), \end{aligned} \tag{1}$$

where f indexes the four latent state factors – normative influence, meta-belief, network structure, and token activation of our applied factorization of the POMDP state set S – and i indexes the states within each factor. Note that each belief-update corresponds to an "aha!" moment for the agent. We observe that DDPM exhibits only a single "aha!" moment early on, while further random rewiring steps produce minimal additional belief updates. By contrast, active inference

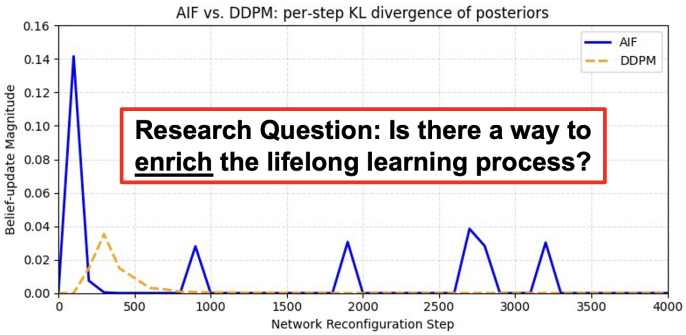

Fig. 3. Active inference (AIF) lifelong learning via "aha!" moments in contrast to DDPM: Mean belief-update magnitude $\Delta_{\mathrm{KL}}(s)$ obtained from 70 simulation runs for each measurement point.

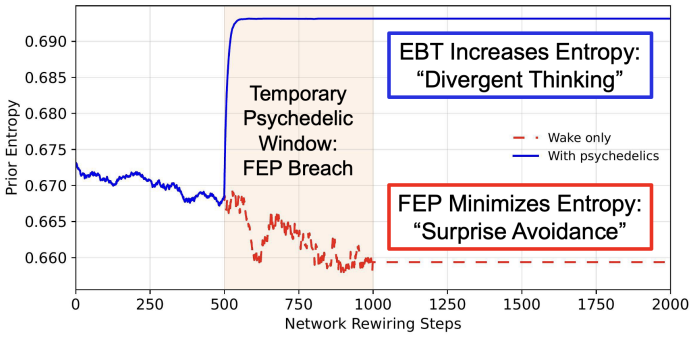

Fig. 4. Entropy of prior belief, captured by POMDP-based generative model parameter $\mathbf{D}$, under the temporary influence of psychedelics: Shaded region represents time window of psychedelic experience.

continually guides the agent to make the most epistemically valuable structural changes throughout the entire lifetime of the rewiring process.

Next, let us enrich the active inference based LL process thru the temporary influence of psychoactives. In [20], Friston *et al.* elaborated on the entropic effect of psychedelics on brain activity by *relaxing the precision of prior beliefs*, i.e., their ascribed importance or salience. Rather than seeking new observations to resolve uncertainty, the agent allows cognitive processes to unfold naturally, enabling holistic perceptions or epiphanies. They proposed a new brain model coined RElaxed Beliefs Under pSychedelics (REBUS) that integrates insights from the FEP with those of the so-called *entropic brain theory (EBT)*. Curiously, psychedelics appear to temporarily breach the FEP by increasing (rather than minimizing) brain entropy. The entropic brain effect of psychedelics increases the complexity of brain connectivity, leading to elevated brain entropy over an expanded, flattened free energy landscape. As a result, psychedelics promote a window of exceptionally high plasticity that favors divergent thinking – a key component of creative thinking and discovering novelty that was previously hidden from clear sight. Insights of this sort are often felt as transcendent in their profundity. Interestingly, as the psychedelic effect begins to subside, the brain settles back into its default regimen of free-energy minimization, but typically will not return entirely as before – a positive residual impact often referred to as the *psychedelic afterglow*. Clearly, the open research challenge is how to map the positive effects of psychoactives to specific parameters of the POMDP-based active inference generative model.

According to [6], precision is the inverse of variance and is closely related to the negative entropy of $Q(s)$:

$$-H[Q(s)] = \mathbb{E}_{Q(s)}[\ln\, Q(s)]. \tag{2}$$

A simple way to parameterize $Q(s)$ and make it more or less precise is to use a Gibbs form with a *precision parameter* ω:

$$Q(s|\omega) = Cat(\sigma(\omega\, \ln\, \mathbf{D})) \tag{3}$$

where σ denotes the softmax function and precision parameter ω multiplies the log prior belief $\mathbf{D}$. Hence, ω may be interpreted as a *neuromodulatory gain-control device*, whereby increasing precision raises the confidence in posterior belief $Q(s)$, and vice versa. Given that precision parameter ω is not part of the open-source Python library pymdp for active inference, we've built a standalone NumPy-based simulation that implements precision effects via Dirichlet sampling. We chose the following two precision parameter settings: (i) *Default*: $\omega = 12$ for high precision during wake periods, and (ii) *Temporary Psychedelic Window*: $\omega = 1.5$ for low precision during psychedelic period.

For illustration, Fig. 4 shows the evolving entropy of prior belief $\mathbf{D}$ during a 500-step long psychedelic window, where we mapped the weakened precision weighting of prior belief $\mathbf{D}$ to a decreased precision parameter ω of the mathematical active inference framework. Further, we realized the psychedelic afterglow effect by deactivating the DMN via suppression of its precision signals to disrupt rigid prior beliefs. The resulting loss of precision produces a free-energy landscape with expanded state space accessibility. Note that Fig. 4 compares two agents, one with psychedelics and one without (wake only). While the awake agent experiences the prior entropy approaching its minimum (FEP regimen), the former one undergoes a sharp increase in prior entropy during the psychedelic window (breached FEP). After precision is restored (step 1000), the prior entropy remains elevated, showing a persistent effect resembling the psychedelic afterglow.

VI. Conclusions

Modern cognitive science has abandoned the outdated assumption that all worthwhile cognitive processes take place only in our ordinary awake state, a flaw called the *single-state fallacy*. Multistate cognitive science recognizes that creative problem solving and exceptional performance take place in other mental states, known as altered states of consciousness, and develops ways of producing these states. Our developed POMDP-based active inference generative model was shown to outperform OpenAI's improved DDPM generative model, rendered embodied and enactive via appropriate modifications, by one order of magnitude due to its superior belief-updating approach and inherent AGI capability of lifelong learning (LL). Further, we enriched the active inference based LL process thru the temporary influence of psychoactives by complementing the FEP with insights from the EBT to temporarily breach the FEP. We showed

that the psychoactive effect of elevated brain entropy over an expanded free energy landscape remains persistently, resulting in something we'd like to call *AI enlightenment* rather than AI embodiment.

Concluding, we note that the dark-room problem is considered a surprisingly serious challenge in the field of cognitive science. While the focus of this paper was on resolving the dark-room problem in active inference, future research may investigate its role in JEPA-style world models and model-based reinforcement learning (RL) models, e.g., POMDP based Dreamer, continual (or lifelong) RL based BC-MDP/ZeUS, and goal-oriented environment inference (GOEI) premised on the concept of redundantly OMDP (ROMDP) using a reduced state space. To our best knowledge, the dark-room problem and use of biomimetic psychoactives have not been yet studied in any of these AI models.

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